

Rajveer Sodhi

(973)382-6195 | Providence, RI | rajveersodhi03@gmail.com | linkedin.com/in/rajveersodhi | github.com/rajveersodhi | rajveersodhi.com

Education

Brown University, Sc.M Computer Science (GPA: 4.00)

2025 - Present

University of British Columbia Okanagan, B.Sc Computer Science Honors (GPA: 4.30)

2021 - 2025

Experience

Software Developer Intern / Avahi, Remote

May 2026 - Sep 2026

- Developed code analysis and transformation modules for Neuron Migrate, a static analysis engine that helps migrate PyTorch/CUDA codebases to AWS Inferentia 2.
- Led the design and architecture of an internal PMO dashboard to replace a manual Jira, Excel, and PowerPoint reporting pipeline with centralized project, revenue, delay, and resourcing views for leadership across 30+ active projects.

Data Analyst Intern / Data POEM, Remote

May 2024 - Sep 2024

- Built scalable connector modules using Mage.ai to interface with PostgreSQL and S3, facilitating the retrieval and transformation of website analytics data from multiple advertisement channels for machine learning model pipelines.
- Focused on data validation and verification to ensure the integrity of ML models, applying analytical skills to identify discrepancies.

Software Developer Intern / UBC Okanagan, Kelowna, BC

Sep 2023 - Dec 2023

- Spearheaded the development of a module within a graphical Decision Support System in Java for long-term degree planning, which is expected to help over 10,000 students plan courses and streamline discussions with academic advisors.
- Added support for over 120 complex degree configurations and used Google CP-SAT Solver to build an optimization model for optimal plan generation, along with various code refactoring and improvements to the optimizer.

Projects

Mossum: Weather Visualization macOS App / SwiftUI, FastAPI, Redis, Render, SwiftData

May 2025 - Present

- Shipped a macOS app that displays live, customizable weather animations on the desktop using SpriteKit and the WeatherKit API.
- Designed a FastAPI backend with Redis caching to reduce latency and external API calls.
- Implemented versioned FAQ synchronization, enabling offline persistence in the app via SwiftData.
- Reached 2,700+ downloads and 30+ weeks in the top weather app charts on the Mac App Store.

FlightBar: Flight Tracking macOS App / SwiftUI, FastAPI, Redis, Render, SwiftData

Feb 2025 - Present

- Built a flight tracker app backed by a FastAPI server that aggregates aviation, airport, airline, and weather data into a unified response.
- Implemented data validation, sanitization, and fallback logic to handle inconsistent data from external APIs.
- Added Redis caching with tiered TTLs (seconds up to days) to balance real-time updates, scheduled data, and long-lived assets.
- Currently with 900+ downloads on the Mac App Store.

Full-Stack E-Commerce Platform / PHP, MySQL, JavaScript, Stripe

Jan 2025 - Apr 2025

- Created a full-stack e-commerce website complete with user authentication, dynamic product listings, persistent carts, a responsive user interface, and order management using PHP and MySQL.
- Integrated Stripe checkout for secure payment flows and order placement.

Full-Stack AI-Powered Budgeting Website / Next.js, Plaid API, Pinecone, PostgreSQL, Docker

Sep 2024 - Apr 2025

- Developed a full-stack budgeting platform to help users visualize and manage spending habits using LLM-driven insights and a user context-aware chatbot powered by the Gemini API, applying software development lifecycle principles for smooth sprints.
- Engineered a vector-based transaction retrieval system by embedding and storing user financial data from Plaid API in Pinecone and PostgreSQL for fast, semantic querying.

Publications

Comparison of Unencumbered Interaction Techniques for Head-Mounted Displays / ACM ISS

Oct 2024

- Published peer-reviewed HCI research with 20+ participants evaluating cutting-edge command selection techniques for head-mounted displays, such as Hand-Proximate User Interface (HPUI) and gaze-tracked input.

Skills

Languages: Swift, Python, JavaScript, TypeScript, Java, SQL

Frontend: React, Next.js, HTML/CSS, Tailwind CSS

Backend and Data: FastAPI, Flask, Node.js, Express.js, PHP, Pandas, NumPy

Databases and Tools: PostgreSQL, MySQL, Redis, Prisma, SQLAlchemy, SwiftData, Git, Docker, AWS